

Clustering

Imports

```
In [151]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.decomposition import PCA
from sklearn import preprocessing
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import MiniBatchKMeans
from sklearn.metrics import adjusted_rand_score
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import MinMaxScaler

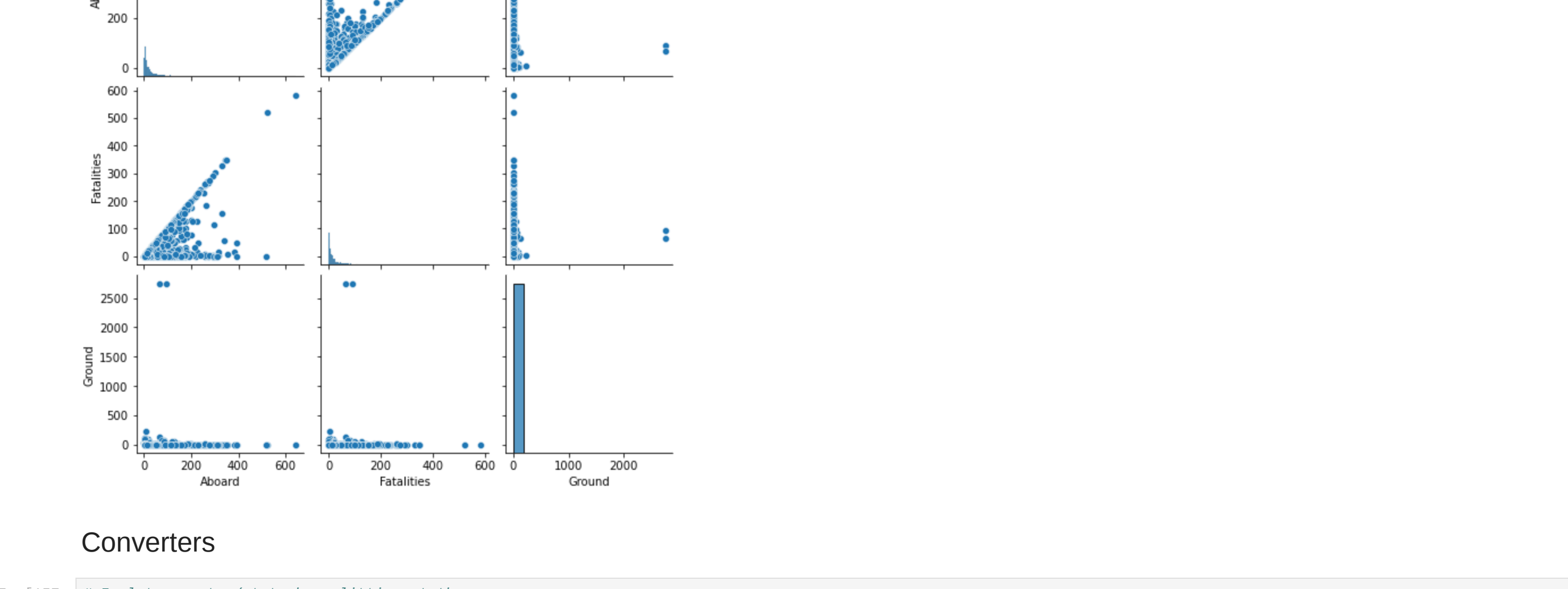
In [152]: df = pd.read_csv('Airplane_Crashes_and_Fatalities_Since_1908.csv')

In [153]: df.head()
```

	Date	Time	Location	Operator	Flight #	Route	Type	Registration	cn/In	Aboard	Fatalities	Ground	Summary	
0	09/17/1908	17:18	Fort Myer, Virginia	Military - U.S. Army	NaN	Demonstration	Wright Flyer III	NaN	NaN	1	2.0	1.0	0.0	During a demonstration flight, a U.S. Army fly...
1	07/12/1912	06:30	AtlantiCity, New Jersey	Military - U.S. Navy	NaN	Test flight	Dingible	NaN	NaN	5.0	5.0	0.0	0.0	First U.S. dingible Akron exploded just offsh...
2	08/06/1913	NaN	Victoria, British Columbia, Canada	Private	-	NaN	Curtiss seaplane	NaN	NaN	1.0	1.0	0.0	0.0	The first fatal airplane accident in Canada oc...
3	09/09/1913	18:30	Over the North Sea	Military - German Navy	NaN	NaN	Zeppelin L-1 (airship)	NaN	NaN	20.0	14.0	0.0	0.0	The airship flew into a thunderstorm and encou...
4	10/17/1913	10:30	Near Johannishai, Germany	Military - German Navy	NaN	NaN	Zeppelin L-2 (airship)	NaN	NaN	30.0	30.0	0.0	0.0	Hydrogen gas which was being vented was sucke...

```
In [154]: df.columns
Out[154]: Index(['Date', 'Time', 'Location', 'Operator', 'Flight #', 'Route', 'Type', 'Registration', 'cn/In', 'Aboard', 'Fatalities', 'Ground', 'Summary'],
      dtype='object')

In [155]: df.dtypes
Out[155]: Date            object
Time            object
Location        object
Operator        object
Flight #        object
Route          object
Type           object
Registration    object
cn/In          object
Aboard         float64
Fatalities     float64
Ground         float64
Summary        object
dtype: object
```



Converters

```
In [157]: # Isolate country/state by splitting at the comma
country=[]
def get_country(location):
    if type(location)==float:
        return "Unknown"
    country=location.split(',')
    return country[len(country)-1].replace(" ", '').lower().replace("off ", '')

In [158]: # Return year
def get_year(date):
    return date.split('/')[2]

In [ ] :
In [ ] :

In [159]: df.Location=df.Location.apply(get_country)
df.Date=df.Date.apply(get_year)
df.Date=df.Date.astype(int)

In [160]: df
```

	Date	Time	Location	Operator	Flight #	Route	Type	Registration	cn/In	Aboard	Fatalities	Ground	Summary	
0	1908	17:18	virginia	Military - U.S. Army	NaN	Demonstration	Wright Flyer III	NaN	NaN	1	2.0	1.0	0.0	During a demonstration flight, a U.S. Army fly...
1	1912	06:30	newjersey	Military - U.S. Navy	NaN	Test flight	Dingible	NaN	NaN	5.0	5.0	0.0	0.0	First U.S. dingible Akron exploded just offsh...
2	1913	NaN	canada	Private	-	NaN	Curtiss seaplane	NaN	NaN	1.0	1.0	0.0	0.0	The first fatal airplane accident in Canada oc...
3	1913	18:30	overthenorthsea	Military - German Navy	NaN	NaN	Zeppelin L-1 (airship)	NaN	NaN	20.0	14.0	0.0	0.0	The airship flew into a thunderstorm and encou...
4	1913	10:30	germany	Military - German Navy	NaN	NaN	Zeppelin L-2 (airship)	NaN	NaN	30.0	30.0	0.0	0.0	Hydrogen gas which was being vented was sucke...
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
5263	2009	06:30	indonesia	Military - Indonesian Air Force	NaN	Jakarta - Maduin	Lockheed C-130 Hercules	A-1325	1982	112.0	98.0	2.0	0.0	While on approach, the military transport cras...
5264	2009	NaN	democratia republicongo	Service Air	NaN	Goma - Isiro	Antonov An-26	9Q-CSA	5005	4.0	4.0	0.0	0.0	The cargo plane crashed while on approach to L...
5265	2009	00:15	brazil	Air France	447	Rio de Janeiro - Paris	Airbus A330-203	F-GZCP	660	228.0	228.0	0.0	0.0	The Airbus went missing over the AtlanticOcean ...
5266	2009	08:30	canada	Strait Air	NaN	Lourdes de BlanSablon - Port Hope Simpson	Briten-Norman BN-2A-27 Islander	C-FJJR	424	1.0	1.0	0.0	0.0	The air ambulance crashed into hills while at...
5267	2009	NaN	india	Military - Indian Air Force	NaN	Mechuka for Jorhat	Antonov An-32	NaN	NaN	13.0	13.0	0.0	0.0	The military transport went missing while en r...

5268 rows x 13 columns

```
In [161]: df['Fatalities'].fillna(0, inplace = True)
df['Aboard'].fillna(0, inplace = True)
df['Ground'].fillna(0, inplace = True)

In [162]: df['Survivors'] = df['Aboard'] - df['Fatalities']
df['Survivors'].fillna(0, inplace = True)

In [163]: df.head()
```

	Date	Time	Location	Operator	Flight #	Route	Type	Registration	cn/In	Aboard	Fatalities	Ground	Summary	Survivors	
0	1908	17:18	virginia	Military - U.S. Army	NaN	Demonstration	Wright Flyer III	NaN	NaN	1	2.0	1.0	0.0	During a demonstration flight, a U.S. Army fly...	1.0
1	1912	06:30	newjersey	Military - U.S. Navy	NaN	Test flight	Dingible	NaN	NaN	5.0	5.0	0.0	First U.S. dingible Akron exploded just offsh...	0.0	
2	1913	NaN	canada	Private	-	NaN	Curtiss seaplane	NaN	NaN	1.0	1.0	0.0	The first fatal airplane accident in Canada oc...	0.0	
3	1913	18:30	overthenorthsea	Military - German Navy	NaN	NaN	Zeppelin L-1 (airship)	NaN	NaN	20.0	14.0	0.0	The airship flew into a thunderstorm and encou...	0.0	
4	1913	10:30	germany	Military - German Navy	NaN	NaN	Zeppelin L-2 (airship)	NaN	NaN	30.0	30.0	0.0	Hydrogen gas which was being vented was sucke...	0.0	

Crash Locations

```
In [164]: # 20 countries/US States with the most crashes near
pd.DataFrame(df.Location.value_counts().sort_values(ascending=False))[:20]
```

Location	
brazil	189
alaska	177
russia	175
canada	149
colombia	149
california	143
france	133
england	105
india	99
china	88
indonesia	87
mexico	83
italy	79
germany	79
australia	78
philippines	68
spain	65
newyork	62
ussr	61
venezuela	58

Operator crash amounts

```
In [165]: df.dropna(subset=['Operator'], inplace=True)
pd.DataFrame(df.Operator.value_counts().sort_values(ascending=False))[:20])
```

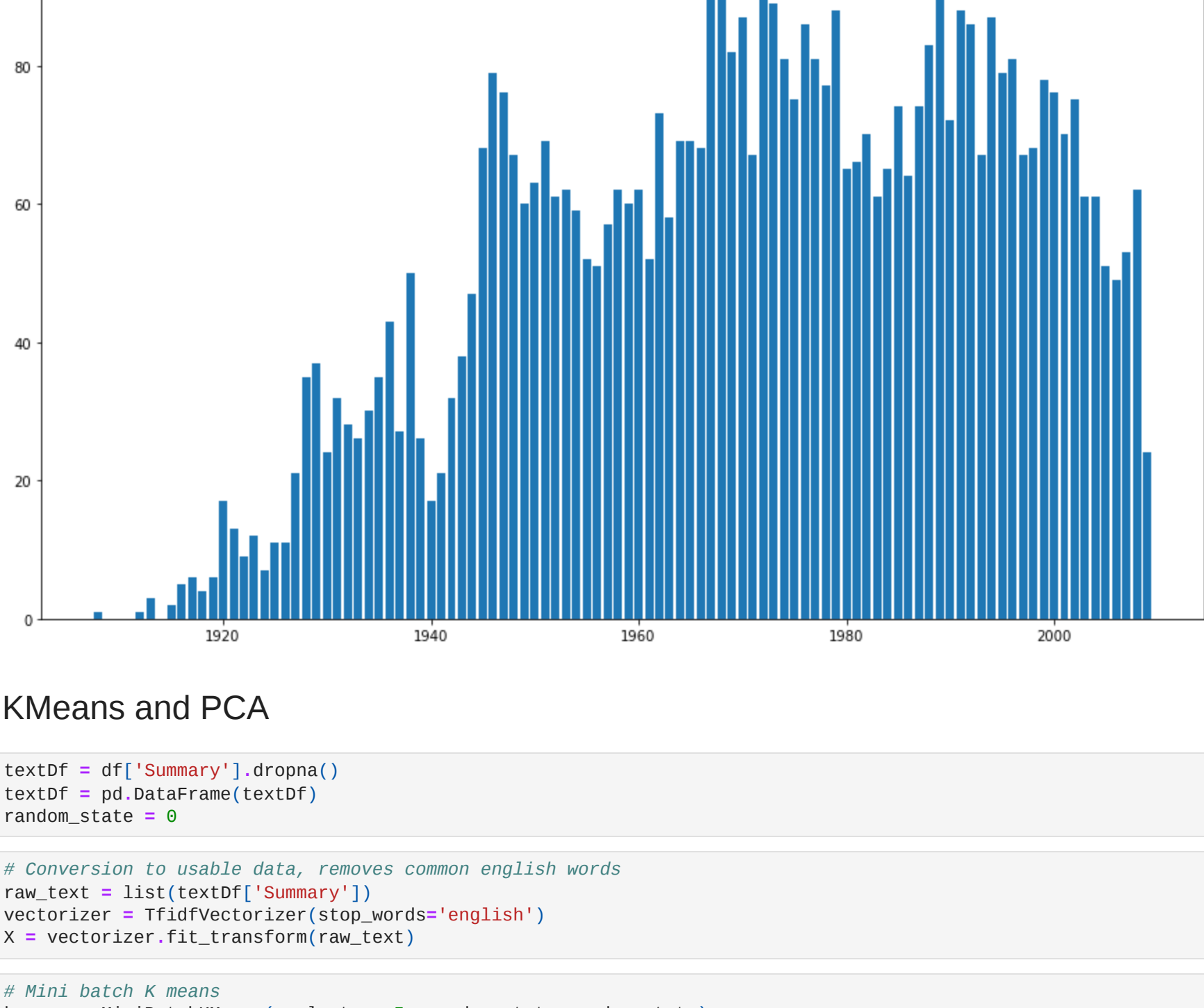
Operator	
Aeroflot	179
Military - U.S. Air Force	176
Air France	70
Deutsche Lufthansa	65
Air Taxi	44
China National Aviation Corporation	44
United Air Lines	44
Military - U.S. Army Air Forces	43
Pan American World Airways	41
Military - U.S. Navy	36
American Airlines	36
US Aerial Mail Service	36
Military - Royal Air Force	36
Indian Airlines	34
Philippine Air Lines	33
KLM Royal Dutch Airlines	33
Private	31
Aerogostale	26
Northwest Orient Airlines	25
Eastern Air Lines	25

Crashes over the years

```
In [172]: pd.DataFrame(df.Date.value_counts().sort_values(ascending=False))[:20]
```

Date	
1972	104
1968	96
1989	95
1967	91
1973	89
1979	88
1991	88
1970	87
1994	87
1976	86
1992	86
1988	83
1969	82
1977	81
1974	81
1996	81
1946	79
1995	79
1999	78
1978	77

```
In [198]: plot = pd.DataFrame(df.Date.value_counts())
plt.figure(figsize = (15, 10))
plt.bar(x=plot.index, height= plot["Date"])
plt.title("Crashes per year")
plt.show()
```



KMeans and PCA

```
In [212]: textDf = df['Summary'].dropna()
textDf = pd.DataFrame(textDf)
random_state = 0

In [213]: # Conversion to usable data, removes common english words
raw_text = list(textDf['Summary'])
vectorizer = TfidfVectorizer(stop_words='english')
X = vectorizer.fit_transform(raw_text)

In [214]: # Mini batch K means
kmeans = MiniBatchKMeans(n_clusters=5, random_state=random_state)
kmeans.fit(X)

Out[214]: MiniBatchKMeans(n_clusters=5, random_state=0)

In [215]: kmeans.predict(X)
kmeans.labels_

Out[215]: array([1, 1, 1, ..., 1, 1, 3], dtype=int32)

In [217]: # PCA and plotting (5 clusters)
pca = PCA(n_components=2, random_state=random_state)
reduced_features = pca.fit_transform(X.toarray())
reduced_cluster_centers = pca.transform(kmeans.cluster_centers_)
plt.scatter(reduced_features[:,0], reduced_features[:,1], c=kmeans.predict(X))
plt.scatter(reduced_cluster_centers[:, 0], reduced_cluster_centers[:,1], markers='x', s=300, c='b')

Out[217]: <matplotlib.collections.PathCollection at 0x7fcdaf685dc0>
```



Commonalities

```
In [218]: common = kmeans.cluster_centers_.argsort()[:, :-1]
words = vectorizer.get_feature_names_out()
for i in range(5):
    print("\n")
    print("Cluster %d:" % i)
    for j in common[i, :15]:
        print ("%s" % words[j]),
    print

Cluster 0:
crashed
takeoff
cargo
plane
shortly
taking
attempting
sea
mountain
land
landing
fog
mountains
ocean
mall

Cluster 1:
crashed
aircraft
approach
flight
crew
pilot
airport
miles
control
ground
landing
altitude
runway
land

Cluster 3:
en
route
crashed
disappeared
mountain
plane
cargo
weather
aircraft
flight
struck
mountainous
mountains
terrain
wreckage

Cluster 4:
weather
mountain
conditions
poor
pilot
struck
crashed
vfr
flight
aircraft
flew
plane
adverse
shot
continued

Cause prediction

In [219]: # Key words can be entered and put into a cluster
# Example key words and where they are clustered to
Y = vectorizer.transform(["hit"])
prediction = kmeans.predict(Y)
print(prediction)

Y = vectorizer.transform(["missile"])
prediction = kmeans.predict(Y)
print(prediction)

Y = vectorizer.transform(["engine"])
prediction = kmeans.predict(Y)
print(prediction)

Y = vectorizer.transform(["weather"])
prediction = kmeans.predict(Y)
print(prediction)

Y = vectorizer.transform(["route"])
prediction = kmeans.predict(Y)
print(prediction)

Y = vectorizer.transform(["landing"])
prediction = kmeans.predict(Y)
print(prediction)

[1]
[1]
[2]
[3]
[2]
```

```
In [ ] :
In [ ] :
```